

Akshith Kalvakota

Akshithkalvakota123@gmail.com

AI/ML intern

VIRTUAL TRY-ON USING TPS

Virtual Try-On (VTON) is a computer vision technique that allows us to digitally try clothes on a person without physically wearing them. It is widely used in online shopping apps where you can see how a shirt, jacket, or dress looks on your own photo.

To make this work properly, the system needs to handle one big challenge: clothes in images are not flat. They have folds, bends, and different body shapes. This is where TPS (Thin Plate Spline) comes in.

What is TPS (Thin Plate Spline)?

TPS is a mathematical method used for warping and transforming images smoothly.

In simple terms:

- Imagine a rubber sheet with points marked on it.
- If you stretch or move some points, the whole sheet bends smoothly.
- TPS does exactly this with images.

Its role in VTON:

- TPS is used to deform (reshape) the clothing image
- So it matches the person's body pose and shape
- Instead of just pasting a shirt on a person, TPS bends the shirt so it fits naturally.

How they both work together?

1) Input Images

Person image (e.g., manstanding.jpg)

Cloth image (e.g., jacket.jpg)

2) Human Parsing / Pose Detection

Detect body parts using MediaPipe.

Estimate pose using keypoints.

This helps the model understand how the person is standing

3) **Cloth Preprocessing**

Remove background from cloth image.

Extract only the garment (using segmentation or rembg).

4) **TPS Warping**

TPS takes:

- Cloth shape
- Body keypoints
- Then it warps the cloth so it matches:
- Body posture
- Body size and alignment

5) **Feature Alignment**

The warped cloth is aligned with the person image.

Ensures sleeves, shoulders, and torso match properly.

6) **Try-On Synthesis (Final output)**

A neural network blends:

- Person image
- Warped cloth
- Removes gaps, overlaps, and artifacts

WORKFLOW:

Person Image + Cloth Image

→ Pose Detection

→ Cloth Segmentation

→ TPS Warping

→ Alignment

→ Image Synthesis

→ Final VTON Output

I tried to create a virtual try-on System using this code:

```
VTON.py > ...
1  import cv2
2  import mediapipe as mp
3  import numpy as np
4  from rembg import remove
5
6  person = cv2.imread("manstanding.jpg")
7  cloth = cv2.imread("jacket.jpg")
8
9  h, w = person.shape[:2]
10 cloth_rgb = cv2.cvtColor(cloth, cv2.COLOR_BGR2RGB)
11 cloth = remove(cloth_rgb)
12 cloth = cv2.cvtColor(cloth, cv2.COLOR_RGBA2BGRA)
13
14 mp_pose = mp.solutions.pose
15 pose = mp_pose.Pose()
16
17 rgb = cv2.cvtColor(person, cv2.COLOR_BGR2RGB)
18 res = pose.process(rgb)
19
20 if not res.pose_landmarks:
21     exit()
22
23 lm = res.pose_landmarks.landmark
24 ls = np.array([lm[11].x * w, lm[11].y * h])
25 rs = np.array([lm[12].x * w, lm[12].y * h])
26 lh = np.array([lm[23].x * w, lm[23].y * h])
27 rh = np.array([lm[24].x * w, lm[24].y * h])
28 nose = np.array([lm[0].x * w, lm[0].y * h])
29 shoulder_width = np.linalg.norm(ls - rs)
30 torso_height = np.linalg.norm((lh + rh) / 2 - ((ls + rs) / 2))
31
32 scale = shoulder_width / cloth.shape[1]
33
34 new_w = int(cloth.shape[1] * scale * 2.3)
35 new_h = int(cloth.shape[0] * scale * 2.6)
36
37 cloth = cv2.resize(cloth, (new_w, new_h))
38 center_x = int((ls[0] + rs[0]) / 2)
39 top_y = int((ls[1] + nose[1]) / 2)
40
41 x1 = center_x - new_w // 2
42 y1 = top_y
```

```
42  y1 = top_y
43
44  x1 = max(0, x1)
45  y1 = max(0, y1)
46
47  x2 = min(x1 + new_w, w)
48  y2 = min(y1 + new_h, h)
49
50  cloth = cloth[y2 - y1, :x2 - x1]
51
52  output = person.copy()
53
54  alpha = cloth[:, :, 3] / 255.0
55
56  for c in range(3):
57      output[y1:y2, x1:x2, c] = (
58          alpha * cloth[:, :, c] +
59          (1 - alpha) * output[y1:y2, x1:x2, c]
60      )
61
62  cv2.imshow("Improved Try-On", output)
63  cv2.waitKey(0)
64  cv2.destroyAllWindows()
```

I used “rembg” which is a Python library used to remove the background of an image. Used it on the cloth image to place it correctly on the person image.

I used these images:



And the output I got is:



ISSUES FACED:

But output is just the overlay of the cloth on the person image because:

It lacks key components like TPS-based warping, human parsing, occlusion handling, and deep learning-based cloth synthesis, which are required to realistically adapt the clothing to the human body. As a result, the cloth does not naturally deform or integrate with the body and instead appears like a sticker placed on top of the person.

There are a lot of things that a basic code can't implement to get a full-on perfect virtual-try on system.

WHAT I LEARNED:

In this task, I learned how to build a basic Virtual Try-On system using OpenCV, MediaPipe, and rembg. I used pose estimation to detect human body keypoints and aligned a clothing image onto a person image, along with background removal, resizing, and blending to simulate a try-on effect. Through this, I understood that while simple rule-based image processing can create a basic output, it is not enough to achieve realistic results. I also realized the main difficulties in Virtual Try-On, such as accurate cloth deformation, body segmentation, occlusion handling, and realistic image synthesis. These require advanced techniques like TPS warping and deep learning-based models, which are not part of my current implementation. Going forward, I plan to explore these concepts in more depth and learn how real AI-based VTON systems are built.